Predicting Biomes Using Random Forest Machine Learning

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1 Introduction

Biome maps are important tools in the study of ecosystems and plant geography. Building on early methods of assigning biomes based on temperature and precipitation, the target biome observations were generated by the global dynamic vegetation model (DGVM) LPJ-GUESS, developed at Lund university. This project trains Random Forest (RF) models to predict discrete classes of biomes using precipitation, radiation, and seasonal temperature. We investigate how well RF models trained only on 1961–1990 seasonal climate statistics reproduce the observed biome map ("Biome_obs") on a global grid. We compare how well our model works to predict the biome identities from a global random 4:1 training to test ratio and when tested on India excluding the region from training data.

The primary objectives are: (i) to evaluate a binary classifier for one biome; (ii) to compare an exhaustive and a culled multiclass classifier on a global random split and on India as a hold-out region; (iii) to explain the predictions with SHAP values; (iv) to emulate LPJ-GUESS net primary production (NPP) and vegetation carbon (VegC) with RF regressors; and (v) to compare the RF predictions with the LPJ-GUESS biome classifications Biome_Cmax and Biome_LAI.

2 Method

2.1 Data

The dataset used for our binary and multiclass classifiers holds 59,191 grid cells representing all locations with observations. For each cell, daily precipitation, temperature (min, mean, max), and shortwave radiation for 1961–1990 are summarized per season as mean, median, and standard deviation, alongside soil observations and LPJ-GUESS output. The observed biomes are imbalanced: the most common biome covers 18.7% of sites and the least common covers 0.2% of sites.

2.2 Variables

All outputs from the DGVM LPJ-GUESS were excluded from the training features, because features such as "Biome_LAI" matches Biome_obs closely for several classes and would turn the classifier into a copy of the model it is compared against. Our chosen target variable "Biome_obs" from LPJ-GUESS was used for the binary and multiclass regressions. LPJ-GUESS output variables "NPP" and "VegC" were each used as targets for a RF regression model. Latitude and longitude were also excluded so that the model learns climate rather than location. This leaves a set of (5 x 4 seasons, mean and standard deviation) total variables in the baseline model, increased by 7 variables when including all the soil data in the exhaustive model, and reduced to (3 x 4 seasons, mean only) variables in the final culled model (Table 1).

Three multiclass feature sets were compared:

1. **Baseline:** Precipitation, temperature max/min/mean, shortwave radiation for all four seasons. Mean and standard deviation inlcuded for all variables.
2. **Exhaustive:** All climate data from baseline model and all soilmap.dat data: “clay”, “pH”, “silt”, “sand”, “orgC”, “CN”, and “cellfraction”.
3. **Culled:** Precipitation, temperature, and shortwave radiation mean values for all four seasons.

2.3 Training and Testing

Every model was tested on a global random 80/20 split (47,352 training / 11,839 test cells) and on a spatial hold-out trained on the 58,087 cells outside India and tested on its 1,104 cells. All RF models used the scikit-learn Python library. The binary classification model predicts moist savanna, a dominant Indian biome which experienced many errors in the multiclass India hold-out models. All multiclass methods predict the complete set of 18 biomes

2.4 Evaluation Metrics

- **Test accuracy** correctly classified cells / total cells.
- **Majority baseline** accuracy when always predicting the most common training biome.
- **Balanced accuracy** mean of each biome’s class recall.
- **F1 score (per-biome)**

$$\mathrm{F1} = \frac{2 \times \mathrm{precision} \times \mathrm{recall}}{\mathrm{precision} + \mathrm{recall}} \quad \text{where:} \quad \mathrm{precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \quad \mathrm{recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{1}$$

- **Macro F1:** the average of the per-biome F1 scores.

Balanced accuracy and macro F1 are the recall and F1 score averaged over all biomes, so rare biomes count as much as common ones. The NPP and VegC regressions are evaluated with the square root of mean squared error (RMSE) in the units of the target, and with R^2.

2.5 SHAP Analysis

SHAP values split the reasons behind the model predictions into contributions from each variable: for one cell, the base value (the average predicted probability of a biome) plus all SHAP values equals the model’s probability for that biome. The model then choses the most likely biome. The SHAP value average size over the test cells ranks the variables. Our group used SHAP values to examine model performance and determine which variables to cull.

2.6 Future Climate

Using a moderate societal response (SSP2-4.5) to climate change, all temperature data was multiplied by values scaled by latitude with an overall effect of land temperature values increasing approximately 2.5°C on average. The precipitation data was also scaled by a latitude-dependent ratio. Radiation values and the standard deviations were unchanged. The culled model was trained on all present-day cells and applied to both 1961-1990 and 2066 climates.

Global temperature increase was scaled with the following method: within 30° of the equator the factor is unchanged, at >30° latitude the factor rises with the square of distance from 30°. At 90° it reaches 2.5 in the Northern Hemisphere and 1.5 in the Southern Hemisphere. This represents polar amplification, which is stronger in the Arctic than the Antarctic. The factor is then divided by its mean over all grid cells.

3 Results and Discussion

3.1 Multiclass Classification

Model	Features	Accuracy		Weighted F1		Macro F1	
		Global	India	Global	India	Global	India
Majority baseline	–	0.186	0.018	–	–	–	–
Baseline	40	0.872	0.439	0.871	0.383	0.778	0.351
Exhaustive	47	**0.877**	0.441	**0.875**	0.378	**0.785**	0.342
Culled	12	0.871	**0.508**	0.869	**0.488**	0.775	**0.383**

Table 1: *Multiclass results on the global random 80/20 split and the India hold-out (trained on all other cells). Best value in each column in bold. The majority baseline always predicts the most common training biome.*

On the global random split, all three models classify about 87 % of test cells correctly, far above the 0.186 majority baseline, with macro F1 0.78–0.79 (Table 1). The high performance comes from near-identical neighbouring cells in the training set. On India in the hold-out model, accuracy falls to 0.44–0.51 and macro F1 to 0.34–0.38, and the binary moist savanna classifier drops from 0.989 to 0.758. The learned biome rules transfer only partly to a new region, and the hold-out model does not benefit from training on the identity of near neighbors. This is especially relevant on the moist savanna border in central India/Deccan plateau region.

Soil gives the best global scores, but only by 0.005–0.007, and did not help in India (macro F1 0.342 against 0.351). The culled set keeps only the seasonal means, and standard deviation correlate strongly with the means. With 12 instead of 40 features, it loses at most 0.003 globally but scores best in India (accuracy +0.07, weighted F1 +0.11, macro F1 +0.03). This is likely because fewer, more general features leave the trees less room to fit patterns that hold only where they were trained.

Most remaining error comes from moist savanna, India's largest biome: the culled model gets only 14% of it right and labels the rest as xeric woodland/shrub (47%), tropical deciduous forest, or dry savanna (18% each). These are neighboring biome types along the rainfall gradient, the errors are in transition zones, without class weights the trees lean toward biomes that are more popular in the training data. All of the models struggled with biome boundaries, particularly for similar forest biomes which was also supported by confusion matrix analyses. This is partly due to a relative lack of distinction between the biomes as outlined in the LPJ-GUESS classifications of Smith et al. (2014).

Tuning did not help: a grid search with 5-fold cross-validation on the non-India cells (scored by accuracy) over maximum tree depth (30–59) and 100 or 150 trees changed the culled model's India macro F1 only from 0.383 to 0.382, so all models kept the defaults (100 trees, no depth limit).

3.2 NPP and VegC Regression

Feature set	NPP (kg C m^{-2} yr^{-1})			VegC (kg C m^{-2})		
	MSE	RMSE	R^2	MSE	RMSE	R^2
Baseline	**0.0125**	**0.1116**	**0.755**	5.114	2.261	0.639
Exhaustive	0.0138	0.1173	0.730	**4.567**	**2.137**	**0.677**
Culled	0.0126	0.1123	0.752	5.675	2.382	0.599

Table 2: *Regression errors on the India hold-out. Best values in bold.*

For NPP, the three models give nearly the same error (RMSE 0.112–0.117, Table 2): baseline and culled perform similarly, and the exhaustive model has 10% higher MSE. VegC behaves

the other way: exhaustive (with soil variables) gives the lowest error and culled gives the highest error. NPP follows the current climate, while VegC builds up over decades which can be represented by soil carbon, so the best feature set depends on the target variable. Errors are largest in mountain regions, where NPP and VegC change sharply over short distances. An RF prediction is an average of training values, so it pulls extremes toward the mean making the mountain training less effective.

3.3 Binary Classification

Moist savanna is approximately 1.4% of total biome sites outside India, the binary classifier trained outside India did not predict moist savanna for any cells in India. If we used class weights in our binary classifier this could have been fixed.

3.4 Future Climate

Using a moderate societal response (SSP2-4.5) to climate change, the land temperature values were increased approximately 2.5°C on average, warming increasing with latitude, and scaled precipitation by a latitude-dependent ratio. Radiation values and the standard deviations were unchanged. The culled model was trained on all present-day cells and applied to both 1961-1990 and 2066 climates. Future predictions were compared with the model's own present-day prediction. Compared to the 1961-1990 data, our culled multiclass model generated cells with different biomes for approximately 20% of total global cells.

4 Conclusion

This report predicted LPJ-GUESS biomes from 1961–1990 seasonal climate with random forests.

1. **Performance:** Multiclass model accuracy is 0.87–0.88 globally but 0.44–0.51 in India, the binary moist savanna model falls from 0.989 to 0.758, so the biome rules transfer only partly to a new region.
2. **Errors:** Most errors are between neighbouring biomes, above all moist savanna, of which only 14% is classified correctly in India.
3. **Variables:** Soil adds at most 0.007 globally and nothing in India, while 12 seasonal means score best in India; tuning did not change this.
4. **Regression:** The RF is a fair surrogate for NPP ($R^2 \approx 0.75$ for all feature sets) and a weaker one for VegC (R^2 0.60–0.68), where soil helps most.
5. **Future climate:** Under SSP2-4.5, about 20% of cells change biome, which shows where climate pushes biomes between classifications.

5 References

Smith, B., Wårlind, D., Arneth, A., Hickler, T., Leadley, P., Siltberg, J., & Zaehle, S. (2014). Implications of incorporating N cycling and N limitations on primary production in an individual-based dynamic vegetation model. *Biogeosciences*, *11*(7), 2027–2054. https://doi.org/10.5194/bg-11-2027-2014

6 Use of AI

In the project Claude Opus 5.5, and Gemini Flash 3.8 were used to create figure drafts, example code for data import, run validations of my ideas using my own code, format .typ documents for the presentation and report, and to give critical feedback on the statistical analyses, model performance, and the wording of my claims in the report. For the future prediction code I used AI to provide articles which helped me arrive at the SSP2-4.5 climate forecast approach and suggest how it could be applied to the existing data.